Throwing Exposure in the Premier Lacrosse League, 2019–2026: A Descriptive Analysis of Publicly Reported Shot and Pass Volume as a Basis for Shoulder-Loading Research

Research draft, kinesiology / biomechanics. Includes modeled throwing load

Abstract

Repetitive overhead throwing is a recognized source of shoulder demand in throwing sports, but public lacrosse data contain no joint kinetics. This paper asks what publicly reported Premier Lacrosse League (PLL) statistics can and cannot say about throwing exposure. We merged eight official season exports (2019–2026; 1,868 player-season rows, 609 players) and analyzed field players with at least three games played (n = 1,173 player-seasons). Mean shot volume was 2.08–2.51 shots per game across seasons. Attackers averaged 5.4 shots and 20.6 passes per game in 2022–2026, versus 0.3 and 5.8 for defenders (Kruskal–Wallis H = 599, $p < 0.001$). We also derive a transparent, speed-squared relative shoulder-load index with an explicit assumption interval. Both measures are exposure estimates, not measurements of tissue damage or injury risk.

1. Introduction

Shoulder loading in throwing athletes depends on repetition, intensity, mechanics and recovery. In lacrosse, laboratory work has compared shooting motion across competition levels: Vincent et al. (2015) studied 39 male players, including six professionals, and reported a professional mean ball speed of 138 ± 7 km/h in an overhead shot task, versus 112 ± 15 km/h in high-school players, with professionals showing about 21% faster crosse angular velocity. That study was a laboratory task and does not describe individual PLL players. The present work therefore addresses a narrower, answerable question: how much throwing-related activity do PLL players publicly record, and how does it vary by season and position?

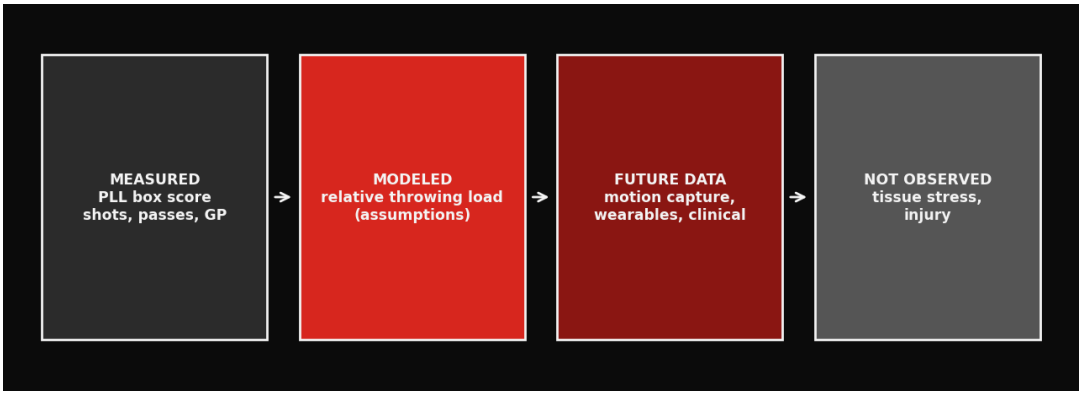


Figure 1. Evidence hierarchy used in this project. Only the first two stages are supported by the data analyzed here.

2. Methods

Data. Eight CSV exports from the official PLL player stats table (stats.premierlacrosseleague.com/player-table), one per season, were merged. Seasons were assigned from team lists and player counts and assumed to be regular-season totals; 2020 is the shortened Championship Series (maximum 7 games). Passes and touches are not recorded for

2019–2021 and were treated as missing for those seasons. Players are identified by name.

Sample. Analyses use field players (attack, midfield, defense, short-stick defensive midfield, long-stick midfield) with at least three games played, excluding goalies and faceoff specialists. **Measures.** Shots per game and, for 2022–2026, passes per game. **Statistics.** Season means with 95% normal-approximation confidence intervals; Kruskal–Wallis test of shots per game across positions; Spearman correlation of shots and passes per game; ordinary least-squares slope across the eight season means. All code and the combined dataset accompany this paper.

3. Results

Season trend. League-wide shot volume per eligible field player was fairly stable (Table 1, Figure 2). The OLS slope across season means was +0.047 shots per game per season (p = 0.026), but with only eight points, a shortened 2020 season and overlapping confidence intervals, this should be read as descriptive, not as a confirmed trend.

Table 1. Shots per game, eligible field players.

Season	n	Mean shots/game	95% CI ±
2019	123	2.25	0.38
2020	128	2.08	0.37
2021	148	2.44	0.38
2022	157	2.39	0.35
2023	153	2.47	0.38
2024	153	2.49	0.37
2025	154	2.51	0.37
2026	157	2.47	0.36

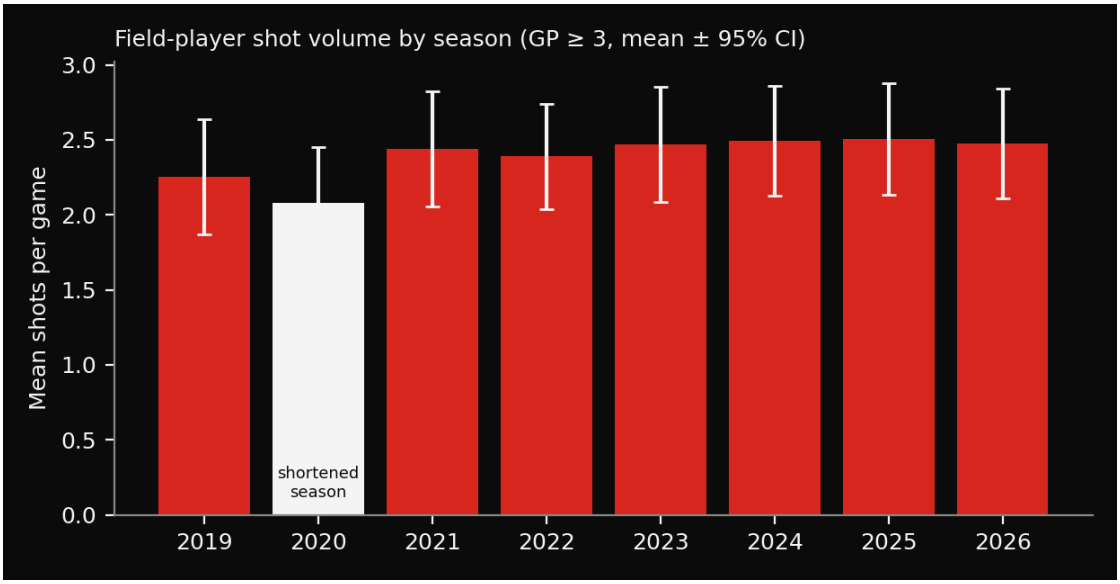


Figure 2. Mean shots per game by season.

Position. Exposure differed sharply by position in 2022–2026 (Figure 3; Kruskal–Wallis H = 599, p < 0.001). Attackers and midfielders carried the highest shot and pass volumes; defenders and long-stick midfielders recorded few shots but a non-trivial number of passes.

Table 2. Mean per-game exposure by position, 2022–2026.

Position	Shots/game	Passes/game
A	5.40	20.56
M	3.70	14.02
D	0.30	5.82
SSDM	0.62	6.15
LSM	0.60	4.13

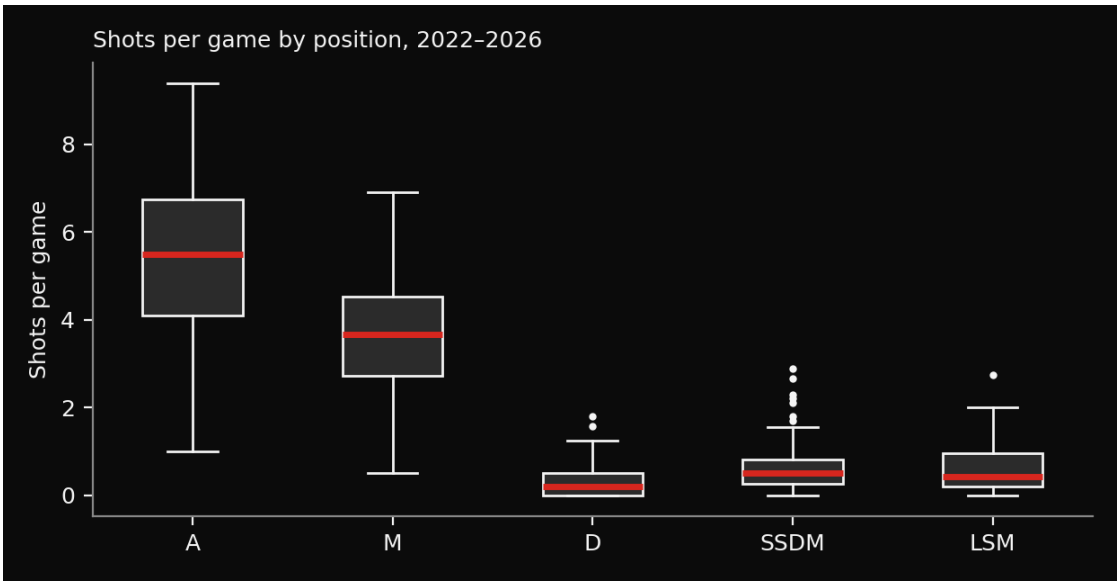


Figure 3. Distribution of shots per game by position, 2022–2026.

Shots and passes. Shots and passes per game were strongly correlated (Spearman ρ = 0.82, n = 774; Figure 4). Much of this reflects position: players who handle the ball most also shoot most. Shots alone therefore understate total throwing, particularly for non-shooters.

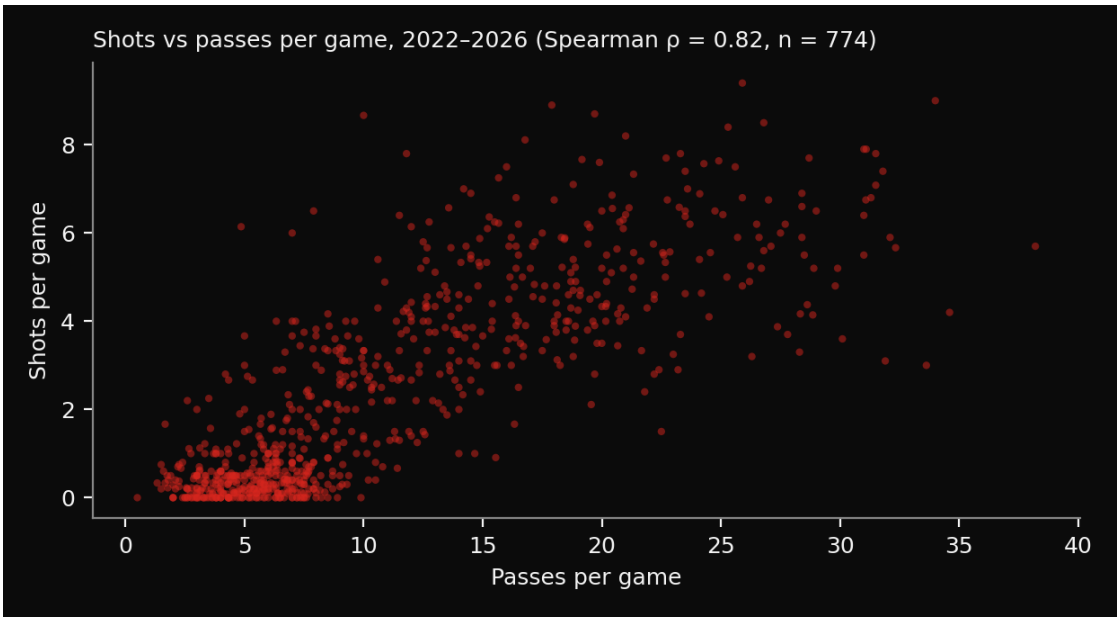


Figure 4. Shots versus passes per game.

Cumulative exposure. Among 143 players with five or more recorded seasons, the largest cumulative recorded shot totals belonged almost entirely to attackers (Figure 5). Because pass counts begin in 2022 and practice and warm-up throws are unrecorded, cumulative totals are lower bounds.

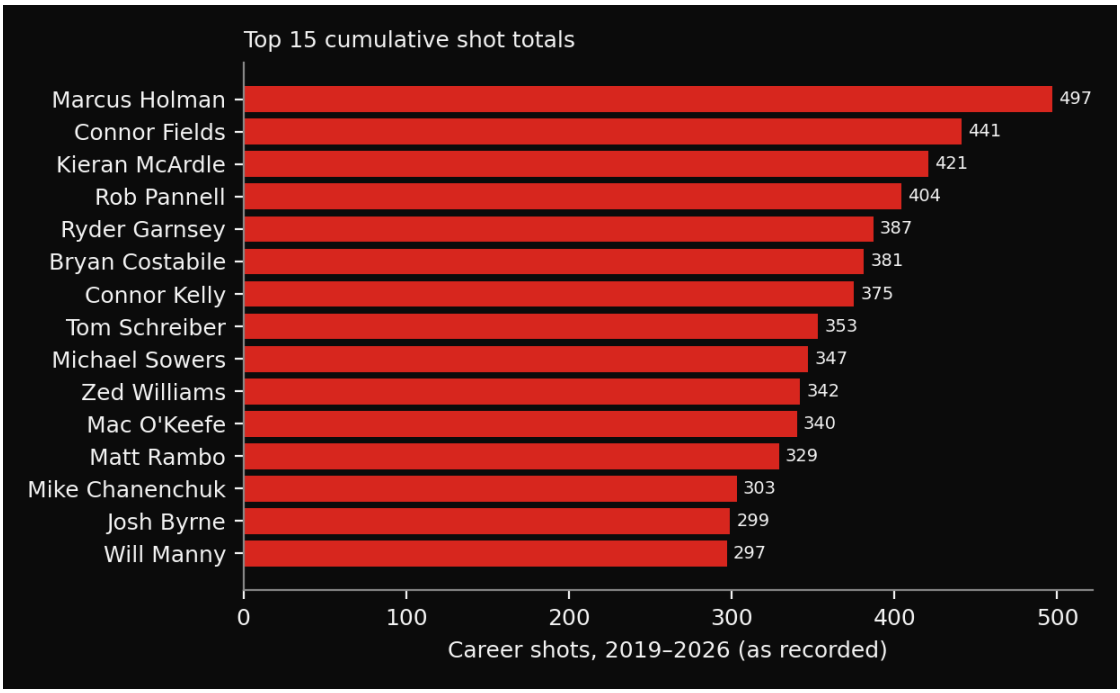


Figure 5. Largest career shot totals in the merged 2019–2026 data.

4. Modeled Shoulder Load

Model. To move from counts toward loading, we built a transparent relative load index. Each throw is weighted by squared ball speed relative to a reference full-effort shot, a kinetic-energy proxy for throwing demand: *load = shots*

$\times 1 + passes \times (v_{pass} / v_{shot})^2$. The reference shot speed is 138 km/h (Vincent et al., 2015, professional mean). Pass speed is **not measured**; we assumed 70 km/h (weight 0.26) and tested 55–85 km/h (weights 0.16–0.38). The unit is the full-effort-shot equivalent. This is a relative exposure index for comparison across players and positions. It is not joules, joint torque, tissue stress or injury probability, and it has not been validated against clinical outcomes.

Findings. At the central weight, attackers carried 10.7 shot equivalents per game versus 1.8 for defenders (Figure 6). Passing contributed about 50% of attackers' modeled load but 83% of defenders' (Figure 7), so ignoring passes would badly understate defensive throwing. Cumulative modeled load for 2022–2026 was highest for Marcus Holman at about 750 shot equivalents (range 596–941 across the pass-speed assumption; Figure 8). Figure 9 places these position differences next to an anatomical schematic of the region repeatedly loaded in overhand throwing.

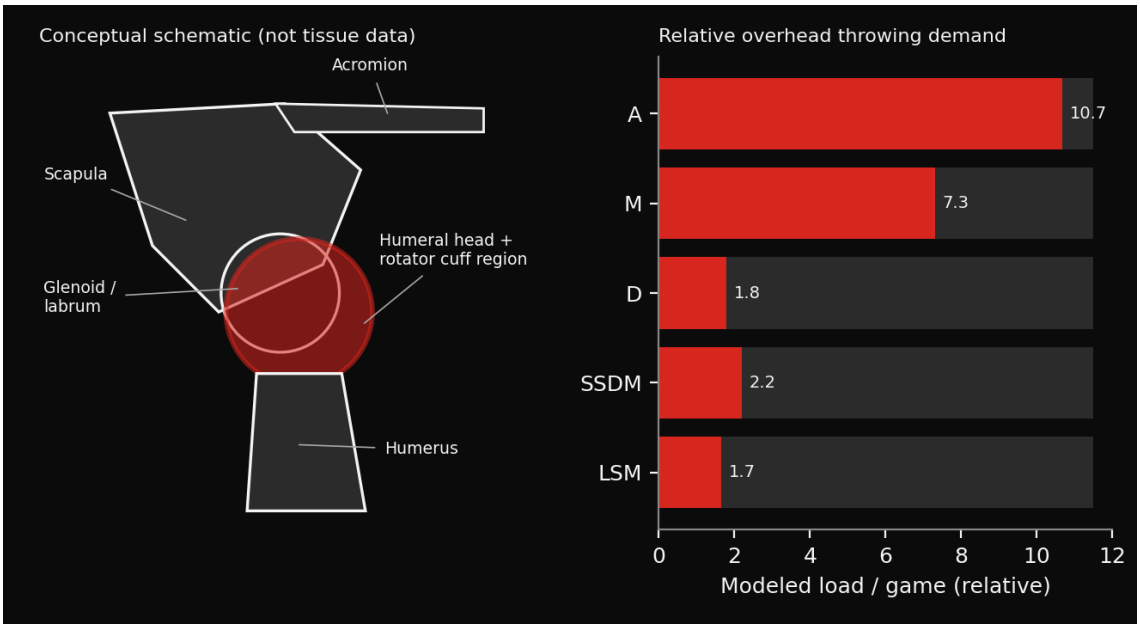


Figure 6. Left: conceptual shoulder schematic; the shaded humeral head and rotator cuff region is illustrative, not measured. Right: central-weight modeled load per game by position.

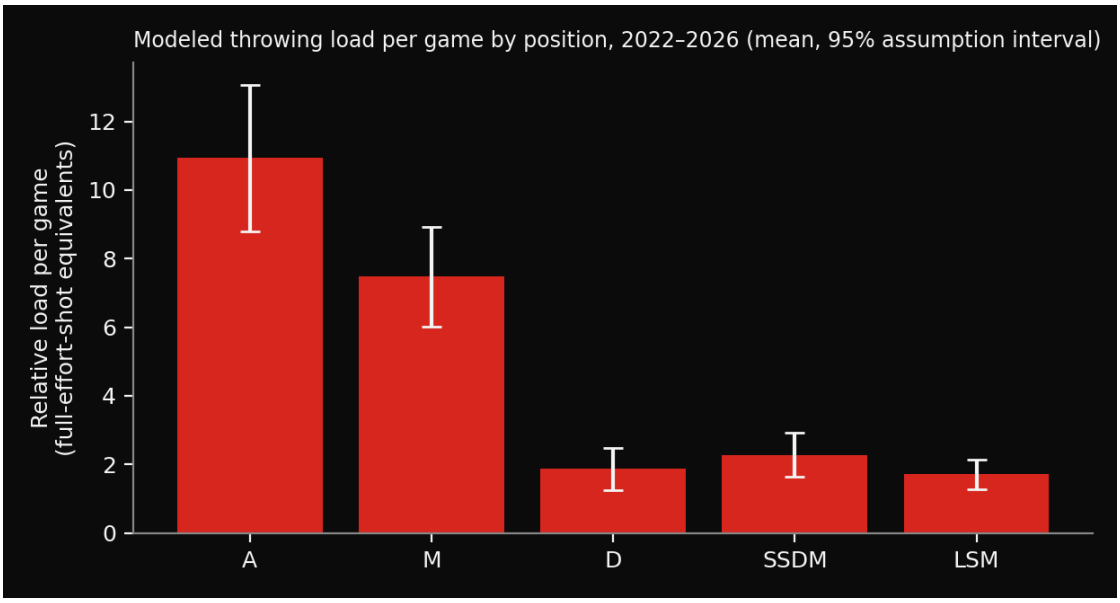


Figure 7. Modeled load per game by position with the assumption interval (95% of 2,000 draws of pass weight).

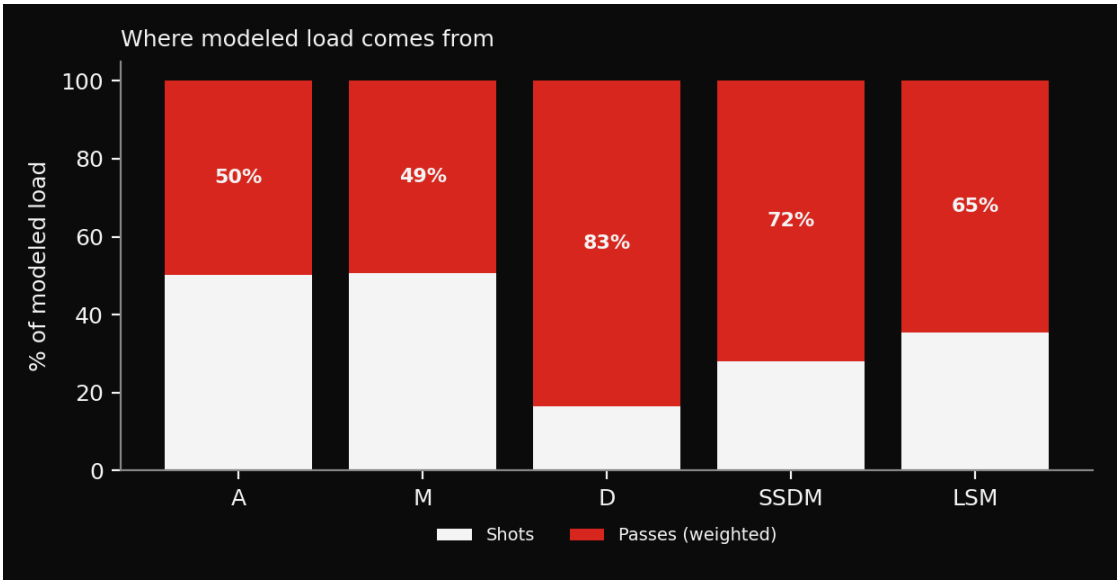


Figure 8. Share of modeled load from shots versus weighted passes, 2022–2026.

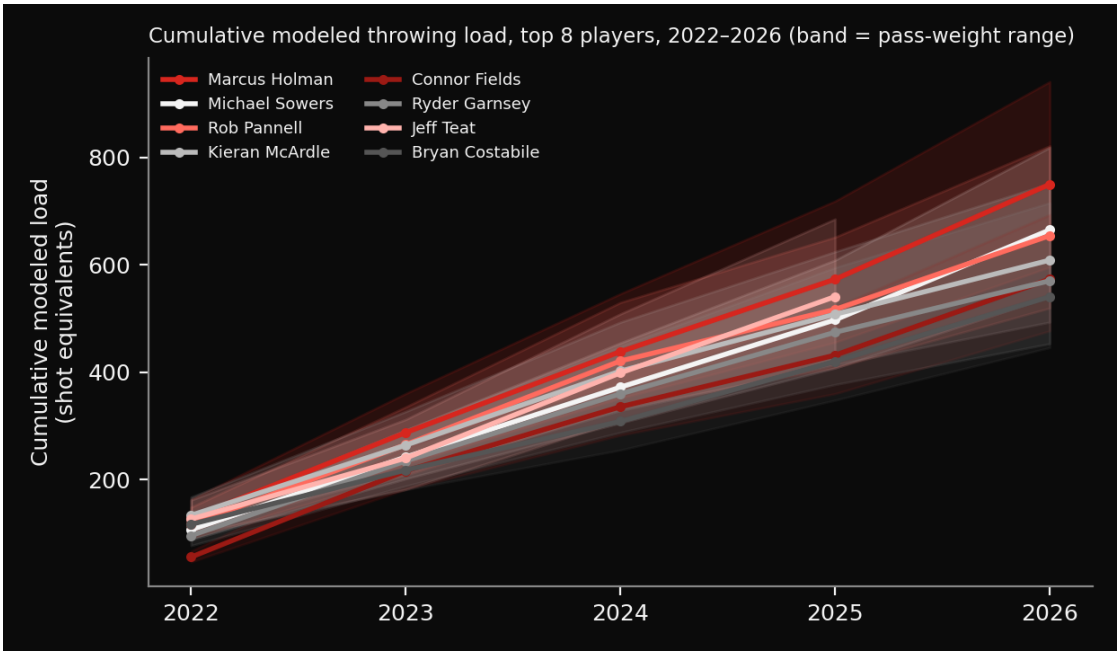


Figure 9. Cumulative modeled load for the eight highest-load players, 2022–2026.

5. Discussion

Publicly reported counts show that exposure is concentrated by role: attackers and midfielders generate most shots and passes, while defenders contribute mainly passing volume. A shoulder-loading study that looked only at shots would miss much of the throwing performed by defensive players. The modeled index shows that role differences shrink but persist once passes are weighted by speed. The next step is to replace the assumed pass speed with measured data from motion capture or wearables, and to compare the index against clinical shoulder outcomes before it is used for decisions about any individual.

6. Limitations

(1) Box-score counts are not joint forces, torque, tissue stress or injury data, and no injury outcome is analyzed; the load index rests on an assumed pass-speed range, ignores throw mechanics, intensity variation, rest and prior injury, and is unvalidated. (2) Practice, warm-up and non-recorded throws are excluded. (3) Pass and touch data are missing for 2019–2021. (4) Season assignment and the regular-season assumption were inferred from file contents and should be verified. (5) Players are matched by name. (6) Literature values come from a laboratory study and must not be attributed to PLL players. (7) Correlations are observational and confounded by position.

References

Premier Lacrosse League. Player Stats Table. https://stats.premierlacrosseleague.com/player-table (season exports downloaded by the project owner).

Vincent HK, et al. (2015). Shooting motion in high school, collegiate, and professional men's lacrosse players. Sports Biomechanics, 14(4), 448–458. doi:10.1080/14763141.2015.1084034.